7 Read the following article and answer the questions that follow.

Anti-reflective coating

When light is incident normally at the interface between a medium with refractive index n_1 and a second medium with refractive index n_2, both reflection and transmission of the light may occur, as shown in Fig. 7.1.

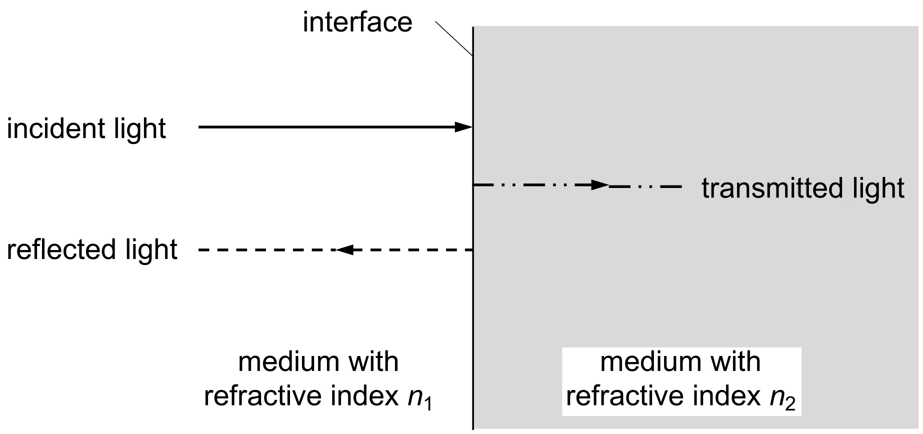


Fig. 7.1

The ratio r of the reflected wave amplitude to the incident wave amplitude is given by

$$r = \frac{n_2 - n_1}{n_2 + n_1}$$

However, these amplitudes can be complex numbers. Therefore, it is more convenient to use the reflectance R and transmittance T where

$$R = \frac{\text{reflected power}}{\text{incident power}}$$

and $R + T = 1$.

Anti-reflective (AR) coating was discovered in 1886 by British Physicist Lord Rayleigh. Upon testing some old, tarnished glass pieces (chemical reactions between the environment and the optical glass of his time tended to cause surface tarnishing on glass as it aged), Lord Rayleigh found that – to his surprise – these tarnished pieces transmitted more light than clean, new pieces.

Today, AR coating is used on eyeglasses to improve vision and reduces eye strain due to the ability of AR coating to reduce reflections from the front and back surfaces of the eyeglass lenses and allow more light to pass through the lenses. Fig. 7.2 shows the reflection of light from the lenses without and with AR coating.

(a) eyeglasses without AR coating **(b)** eyeglasses with AR coating

Fig. 7.2

AR coating is especially beneficial when used on high refractive index lenses, which reflect more light than regular plastic lenses. Generally, the higher the refraction index of the lens material, the more light that will be reflected from the surface of the lens. This can be particularly troublesome in low-light conditions, such as when driving at night.

The reflection of a monochromatic light incident normally on a AR coated lens is minimised when the thickness of the coating is about a quarter of the wavelength, as shown in Fig. 7.3.

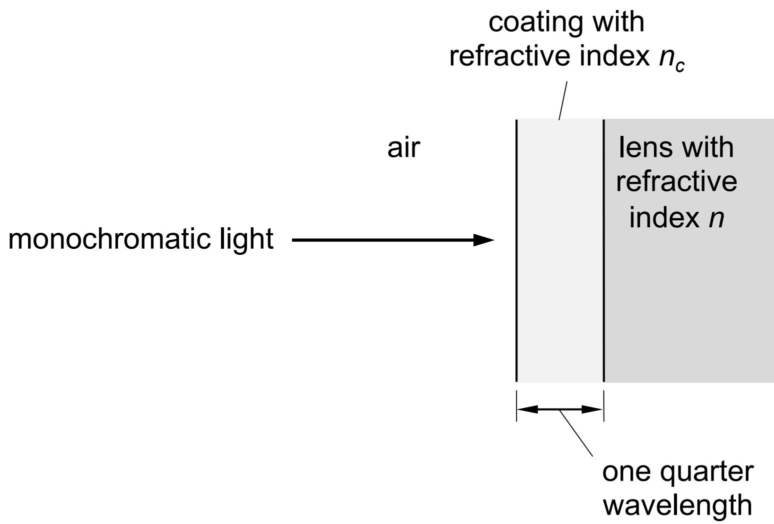
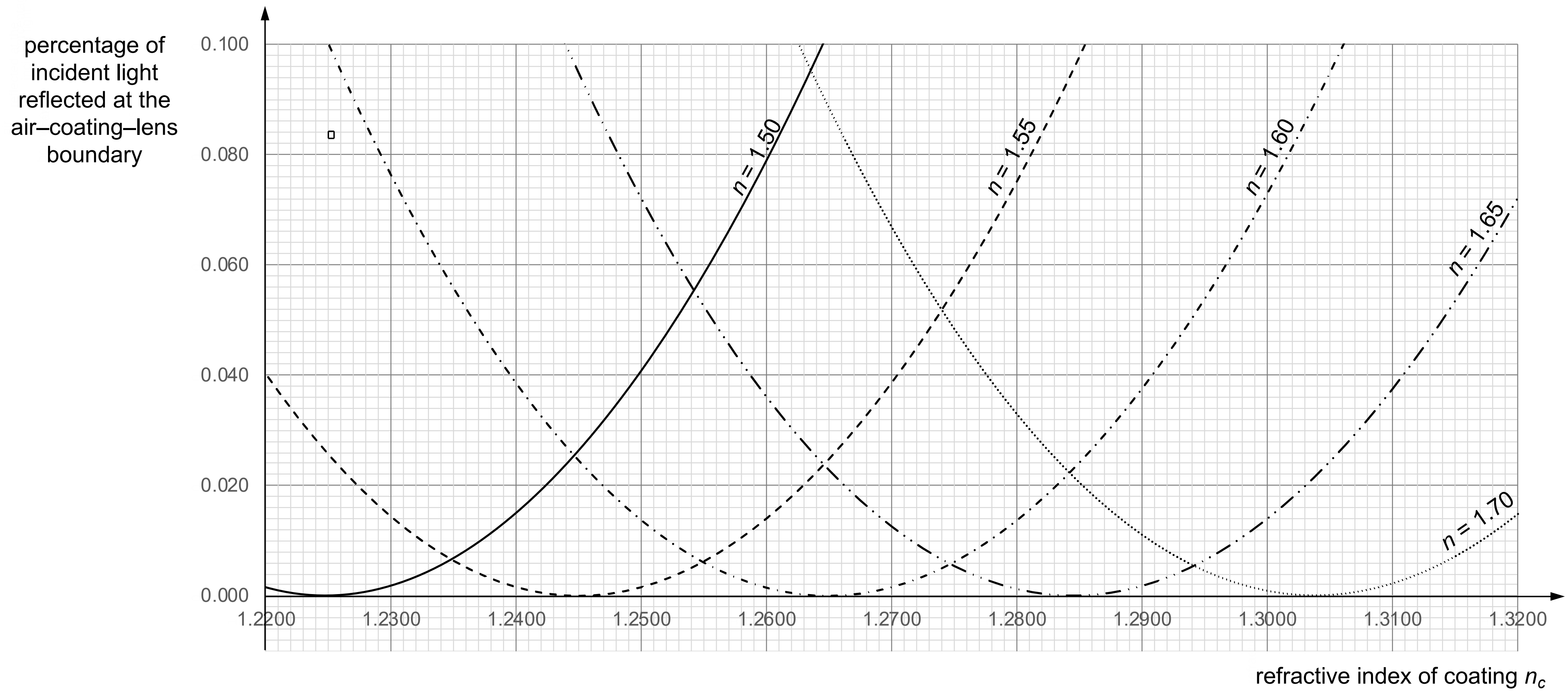


Fig. 7.3

Fig. 7.4 shows how the variation with the refractive index of the coating n_c of the percentage of light reflected at the air–coating–lens interfaces. The optimum refractive index n_c of the coating that minimises the reflection of light for different values of refractive index of lenses can obtained from Fig. 7.4.

Multiple layers of AR coatings on modern lens can virtually eliminate the reflection of visible light from eyeglass lenses, allowing 99.5 percent of available light to pass through the lenses and enter the eye for good vision.



percentage of incident light reflected at the air–coating–lens boundary
0.100
0.080
0.060
0.040
0.020
0.000
1.2200
1.2300
1.2400
1.2500
1.2600
1.2700
1.2800
1.2900
1.3000
1.3100
1.3200
refractive index of coating n_c
n = 1.50
n = 1.55
n = 1.60
n = 1.65
n = 1.70

Fig. 7.4

(a) **(i)** Show that $R=\left(\dfrac{n_2-n_1}{n_2+n_1}\right)^2$

[2]

(ii) **1.** Explain why the sum of R and T is one.

....

.... [1]

2. Deduce the expression for the transmittance T, in terms of n_1 and n_2.

[1]

(iii) Fig. 7.5 shows monochromatic light is incident normally on a lens without AR coating.

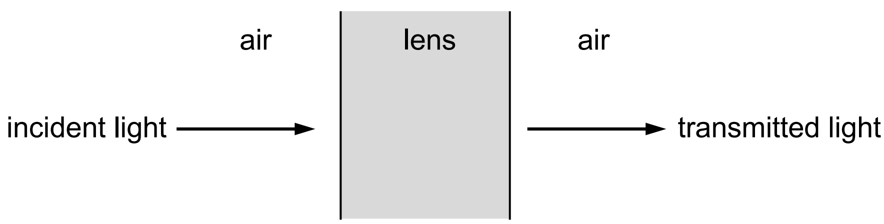


Fig. 7.5

The refractive index of air is 1.00. The refractive indices of two different lens materials are:

	CR–39	polycarbonate
refractive index	1.50	1.58

1. For CR–39 lens, show that about 92% the incident light is transmitted.

[2]

2. Determine the percentage decrease in the transmitted light for a polycarbonate lens.

percentage decrease = ... % [2]

(b) **(i)** When light wave is reflected from the surface of a medium with higher refractive index than that of the medium in which they are travelling, the reflected wave will change phase by 180° or π radians.

Explain why the reflection of a monochromatic light is minimised when the thickness of the AR coating is a quarter of the wavelength.

....

....

....

....

.... [3]

(ii) Fig. 7.6 shows just the incident wave at one instant on the lens with the AR coating.

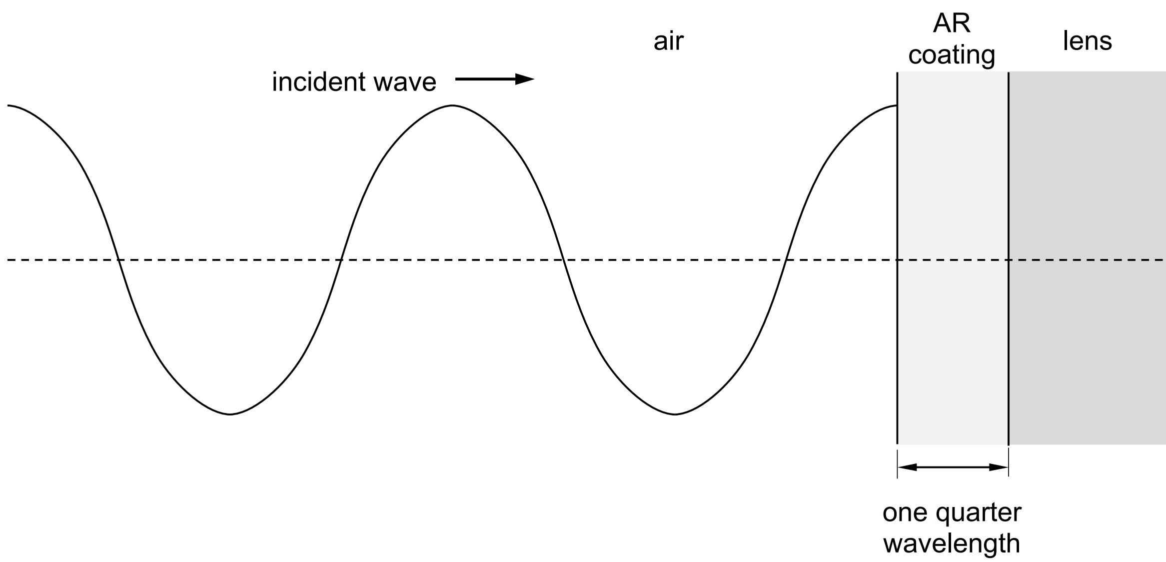


Fig. 7.6

On Fig. 7.6, sketch

1. the reflected wave from the air–coating interface and label it as "C",

2. the reflected wave from the coating–lens interface and label it as "L". [3]

(c) **(i)** Use data from Fig. 7.4 to complete the table below.

refractive index n of lens	optimum refractive index n_c of the coating
1.50	1.2250
1.55	
1.60	
1.65	
1.70	

[1]

(ii) Without drawing a graph, use your data in **(c)(i)** to verify that n_c is proportional to the square root of n.

[2]

(iii) Determine the optimum refractive index of the coating for polycarbonate lens.

optimum refractive index = ... [1]

(d) **(i)** Suggest why there are multiple layers of AR coatings on modern lens.

....

....

.... [2]

(ii) Some high-quality lenses have as many as seven layers of AR coatings.

Estimate the thickness of the AR coatings.

thickness = ... m [2]

[Total: 22]

[Turn over